Module 1 — Core Skills Drill

Assignment type: TalentLMS Assignment (standalone — not embedded in the reading) **Available:** After the Key Concept Session (same evening) **Due:** Same night as the concept session **Submission:** GitHub PR URL — text field in TalentLMS Unit 3 **Points:** 10

Overview

This drill moves you from reading about Python environments and Git workflows to actually executing them from memory. You practiced these commands in Pre-Work; this drill confirms you can reproduce them in a project context — with a venv active, packages installed, and a branch pushed to GitHub.

An autograder runs on your pull request and checks your work. All 3 checks must be green before you submit the PR URL to TalentLMS.

Before You Start

Confirm you have completed:

- [ ] The Module 1 reading
- [ ] The Key Concept Session (attend live or watch the recording in TalentLMS Unit 2)
- [ ] Your GitHub HTTPS credentials are configured (PAT set up from Pre-Work)

Step 1: Accept the M1-D1-git-workflows assignment to get your personal drill repo, then come back here.

Drill Tasks

Task 1 — Clone Your Drill Repo and Set Up the Environment

1. Clone your accepted repo (replace `[username]` with your GitHub username):

```
git clone https://github.com/LevelUp-Applied-AI/m1-d1-git-workflows-[username
cd m1-d1-git-workflows-[username]
```

2. Create a virtual environment named `.venv`:

```
python -m venv .venv
```

3. Activate it (use the command for your OS — see the activation table in the reading, Section 2.1):

```
# Mac / Linux
source .venv/bin/activate

# Windows Git Bash
source .venv/Scripts/activate

# Windows CMD
.venv\Scripts\activate.bat

# Windows PowerShell
.venv\Scripts\Activate.ps1
```

Confirm the `(.venv)` prefix appears in your prompt before continuing.

4. Install the requirements:

```
pip install -r requirements.txt
```

5. Run the environment check:

```
python -c "import pandas; import matplotlib; print('Environment OK')"
```

Expected output:

```
Environment OK
```

No error messages should appear. If you see `ModuleNotFoundError`, the venv is not active or the install did not complete (see Troubleshooting below).

Task 2 — Feature Branch, Commit, and Pull Request

1. Create a branch using your GitHub username:

```
git checkout -b drill/<your-github-username>
```

Replace `<your-github-username>` with your actual GitHub username (e.g., `drill/jsmith`).

2. Open `hello.py` and add one line at the bottom:

```
# Drill completed by Your Name
```

Replace `Your Name` with your actual name.

3. Stage, commit, and push:

```
git add hello.py
git commit -m "Add drill completion note"
git push --set-upstream origin drill/<your-github-username>
```

4. Go to the repo on GitHub and open a Pull Request from your branch to `main`. The PR title can be: `Drill 1 — [Your Name]`.

Task 3 — Verify Git History

Still in your repo, run:

```
git log --oneline
```

Confirm at least two commits appear: the initial starter commit and your drill commit.

Autograder

When you open the pull request, GitHub Actions runs 3 checks automatically:

Check	What it verifies
`test_hello_py_has_completion_note`	`hello.py` contains your completion note (not the placeholder)
`test_environment_imports`	`pandas` and `matplotlib` install and import correctly
`test_branch_name_not_placeholder`	Your branch name contains your actual username, not `<your-github-username>`

All 3 must show ✅ before you submit. If a check is red, read the error message in the Actions tab — it will tell you exactly what to fix.

Submission

Once all 3 autograder checks are green:

1. Copy the URL of your open pull request (e.g., `https://github.com/LevelUp-Applied-AI/m1-d1-git-workflows-jsmith/pull/1`)
2. Paste it into TalentLMS → Module 1 → Core Skills Drill (Unit 3)

Do not close the pull request before submitting.

Troubleshooting

`test_hello_py_has_completion_note` fails: Open `hello.py` — the line must say `# Drill completed by Your Name` with your actual name replacing "Your Name". The literal text "Your Name" will fail.

`test_environment_imports` fails: The `requirements.txt` may have been modified, or the CI environment has an issue. Confirm your local version of `requirements.txt` is unchanged from the original (contains `pandas` and `matplotlib`). Commit any fix and push — the autograder reruns automatically.

`test_branch_name_not_placeholder` fails: Your branch name contains `<` or `>` literally. Delete the branch and recreate it with your actual username: `git checkout -b drill/jsmith` (use your real username).

`ModuleNotFoundError: No module named 'pandas'` locally: The venv is not active. Confirm `(.venv)` is in your prompt. If not, re-run the activation command. Then re-run `pip install -r requirements.txt`.

`Activate.ps1` blocked by PowerShell security policy:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

Then try `Activate.ps1` again.

`git push` rejected: Confirm your remote with `git remote -v`. If you cloned with HTTPS, confirm your PAT is configured.

If you are stuck after trying these steps, post in the course channel before the submission deadline.
